CAPSTONE · BIOMEDICAL ENGINEERING

Computational Assessment of Oxygen Delivery in Decellularized Leaf Vascular Scaffolds

Pipeline development, GreensV4 failure analysis, and a graph-based oxygen-screening prototype

Chris Dargham

May 2026

Why Plant Scaffolds, and Why Oxygen?

The oxygen bottleneck

Mammalian cells consume O_2 continuously; insufficient delivery → hypoxia, growth arrest, death

Diffusion alone limits viable tissue to ~100–200 µm from a vessel *(Krogh, 1919; Carreau et al., 2011)*

Decellularized plant scaffolds

Naturally branched, hierarchical, reticulated vascular templates

Demonstrated for mammalian cell culture: spinach, parsley, apple *(Modulevsky et al., 2014; Gershlak et al., 2017; Dikici et al., 2019)*

Structural preservation alone does not prove the scaffold can deliver O_2

The gap this project targets

A reproducible, image-driven computational pipeline that connects leaf venation geometry → perfusion → oxygen delivery, so candidate scaffolds can be screened before wet-lab seeding.

Literature Review — Two Converging Threads

Plant scaffolds for tissue engineering

Apple scaffolds for 3D mammalian culture — *Modulevsky et al., 2014*

Crossing kingdoms — perfusable spinach scaffolds — *Gershlak et al., 2017*

Decellularized spinach for vascular tissue engineering — *Dikici et al., 2019*

Reticulation = redundancy and damage tolerance — *Katifori et al., 2010; Sack & Scoffoni, 2013*

Microvascular oxygen transport models

Krogh cylinder — capillary-spacing & diffusion limit — *Krogh, 1919*

Green's-function method for O_2 in microvascular networks — *Hsu & Secomb, 1989; Secomb et al., 2004*

Pressure–flow networks (Poiseuille + Kirchhoff) — *Pries et al., 1990; Pries & Secomb, 2008*

Michaelis–Menten O_2 uptake in 3D constructs — *Casciari et al., 1992; Magliaro et al., 2019*

These two literatures have not been formally bridged for reticulated plant venation — the project's working hypothesis is that they can be.

OBJECTIVE

Original Plan: An End-to-End Pipeline

1

Segment

LeafVeinCNN

Image → vein mask, skeleton, calibrated widths

2

Solve flow

NetFlowV2

Graph hemodynamics: Poiseuille + Kirchhoff

3

Solve oxygen

GreensV4

Green's-function intravascular + tissue O_2

Outcome: Stages 1 & 2 succeeded. Stage 3 failed — and the failure became the central methodological finding.

PIPELINE

Final Workflow — Five Stages

STAGE 1

Image segmentation

LeafVeinCNN

Mask + skeleton + widths

STAGE 2

Graph preparation

MAT → Network.dat

Solver-ready graph

STAGE 3

Network hemodynamics

NetFlowV2

Pressures + flows

STAGE 4

Oxygen solver test

GreensV4

Failed — diagnosed

STAGE 5

Oxygen screening

OxySim (custom)

pO_2 maps + metrics

Reproducible

image → flow → graph pipeline

Documented

GreensV4 domain-of-validity failure

Implemented

OxySim screening prototype

Stage 1 — Segmenting the Vascular Network

Processing settings

Setting	Value
Image source	Zenodo CHAMBASA cleared-leaf dataset
Calibration	9.62 µm / px (fixed)
Image rescale	2× before CNN
Ensemble	Random, 3 models averaged
Outputs kept	Full-width mask, skeleton, widths, MAT export

Methodological reasoning

Cleared-leaf prep exposes venation cleanly for skeletonization (Blonder et al., 2025).

Calibration is fixed because lengths, radii, and resistance all scale from it — Poiseuille $G \propto r^4$.

2× rescale improves CNN sampling of fine veins; it does not add biological resolution.

Ensemble of 3 lowers single-model selection bias (Fricker, Blonder & Xu, 2021; Xu et al., 2021).

Two products feed downstream: full-width mask (vessel area) + skeleton (graph topology).

Stage 2 — From Image to Solver-Ready Graph

Polylines → segments

Centrelines resampled into short straight NetFlowV2 segments — preserves arc length so conductance isn't overestimated.

Diameter assignment

Calibrated widthCalibrated when available; fallback = 2× distance transform of full-width mask. Min-diameter floor applied.

Boundary validation

Every BC node forced to a degree-1 terminal post-discretization — interior BCs are auto-relocated to the nearest valid terminal.

Repair & QC

Cap extreme diameters, remove sub-0.5 µm artefacts, classify components, preserve unperfused fragments — full repair log retained.

Why this stage matters: Poiseuille conductance scales as $G \propto r^4$ — small radius errors produce large flow errors (Pries & Secomb, 2008).

Stage 3 — NetFlowV2 Solves the Hemodynamics

Per-segment Poiseuille relation

$$Q_e = G_e\,(P_i - P_j), \quad G_e \propto r^4 / (\mu\, L_e)$$

Interior nodes satisfy Kirchhoff conservation of volumetric flow → global linear system. Reticulated loops are fine here: they are handled by the global solve.

Demonstration network — mass balance OK

Metric	Value
Total segments	1,130
Active (non-zero flow)	1,130
Total nodes	972
Boundary nodes	36 (31 inlet, 5 outlet)
Boundary inflow	776.162 nL/min
Boundary outflow	776.162 nL/min
Balance error	**3.0×10^{-6} nL/min**

NetFlowV2 stays valid for the rest of the project — the failure that follows is at the oxygen stage, not the hemodynamic stage.

NEGATIVE RESULT

Stage 4 — GreensV4 Fails on Plant Venation

Two independent failure modes — neither is an implementation bug

Both stem from a domain-of-validity mismatch between Green's-function microvascular solvers and reticulated leaf networks.

FAILURE 1

Memory scaling

Dense vessel-vessel matrices scale as $O(n_v^2)$

Test case: 61,021 vessel sources × 98,368 tissue points → a single dense float matrix ≈ **14.9 GB**

FAILURE 2

Topological ordering

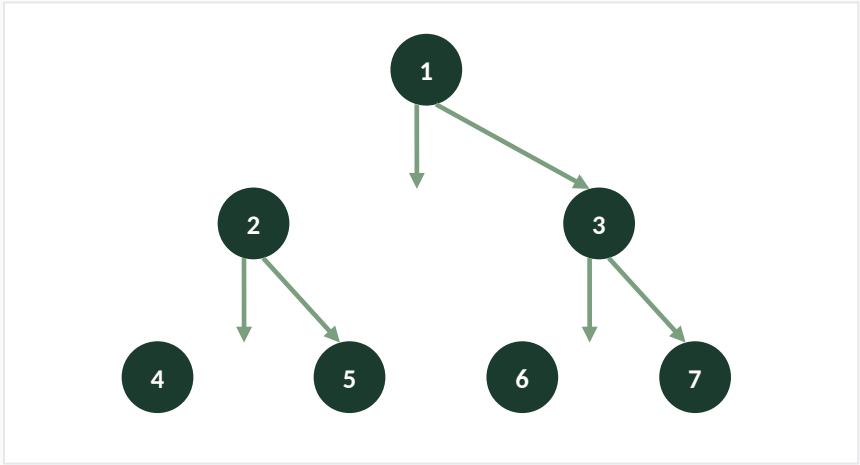
Convective O_2 pass needs a one-way upstream→downstream rank.

Reticulated leaves contain **loops + near-zero flow segments** → dependency graph is **non-rankable.**

FAILURE 2 — IN DEPTH

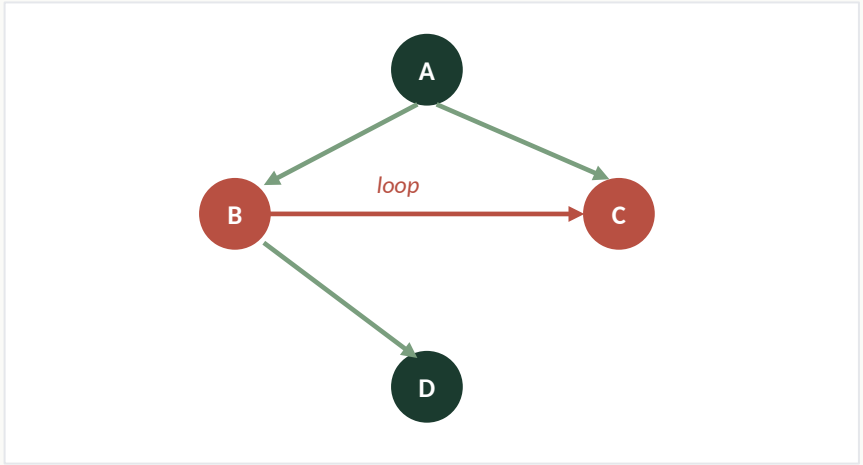
Why Reticulation Breaks the Convective Pass

Vascular tree (microvasculature)



✔ *Rankable: every node processed after all upstream inflow.*

Reticulated leaf venation (loops)



✖ *Loop B ↔ C means neither can be processed first → solver halts.*

Biology: those loops are exactly why leaves are robust to damage and fluctuating demand (Katifori et al., 2010; Sack & Scoffoni, 2013). Pruning them to satisfy GreensV4 destroys the feature under study.

FAILURE DIAGNOSIS

Parameter Tuning Cannot Fix a Topology Failure

Tuning that does NOT help

- Wall permeability (P_wall)
- Tissue diffusivity D
- Consumption Vmax, Km
- Hill P50, exponent n
- Inlet pO_2
- Effective exchange clamp

*These act **after** the convective pass has a valid order. They cannot create one.*

Pruning loops is not a fix

Removing low-flow segments to force a tree changes the model.

What you'd be solving:

an artificial tree extracted from a leaf — not the reticulated scaffold under study.

Conclusion: GreensV4 is operating outside its reliable domain for full-resolution reticulated plant networks.

CUSTOM SOLVER

OxySim — A Topology-Compatible Alternative

Goal: deliver oxygen-accessibility screening on the reticulated graph — without requiring a downstream rank.

2-D tissue grid

Explicit finite-difference reaction–diffusion field with Michaelis–Menten uptake

Skeleton hemodynamics

Sparse Poiseuille pressure-flow on the pixel skeleton (no dense Green matrix)

Lumen O_2 transport

Geodesic upstream propagation along the skeleton with radius-aware wall loss

Screening outputs

pO_2 maps, threshold area fractions, distance-to-vessel CDFs

Claim limit: a screening prototype, not a validated physiological model — outputs are conditional on swept parameters.

How We Represent the Tissue

Governing tissue oxygen equation

$$\partial C/\partial t = D\,\nabla^2 C - V_{eff}\cdot C/(K_m + C) + k_{eff}(x,y)\cdot[\,C_{lumen}(x,y) - C(x,y)\,]$$

$D\,\nabla^2 C$

Fickian diffusion

Truskey et al., 2009

$V{\cdot}C/(K_m{+}C)$

Michaelis–Menten consumption

Casciari 1992; Magliaro 2019

$k(C_lumen - C)$

Vessel→tissue exchange

Hsu & Secomb, 1989

Numerical scheme

- Cartesian grid (384 × 384 default), explicit finite-difference update
- No-flux Neumann boundaries; non-negativity clamp; CFL guard on Δt (Strikwerda, 2004)
- Vessel pixels excluded from consumption (Vmax = 0 inside the vessel mask)

Vessel Wall Exchange & Internal Hemodynamics

Radius-aware wall exchange

$$k(x,y) \approx 2 \cdot P_{wall} / r(x,y)$$

2/r = surface-to-volume ratio of a cylinder

P_{wall} = effective wall permeability (swept, not measured)

Radius from widthCalibrated; fallback = distance transform

Stability clamp $\Delta t \cdot k \leq 0.9$ (numerical safeguard)

Skeleton hemodynamics

$$G = \pi r^4 / (8 \mu \cdot ds)$$

Each skeleton pixel = node, 4-connected to neighbours

Sparse linear system A·p = b; BCs at inlet/outlet pixels

Pressure → edge flow → nodal speed u(x,y)

Velocity floor u_{min} avoids zero-division in low-flow regions

Reachability via BFS from inlet — only reachable skeleton pixels can carry O_2

Lumen Oxygen Transport & Coupled Loop

Lumen O_2 propagation

$$\lambda = k \cdot ds / u$$
$$C_{tot,new} = C_{tot,up} - \lambda \cdot \max(C_{diss,up} - C_{tissue}, 0)$$

Geodesic distance from inlet picks one upstream neighbour per pixel — sidesteps the GreensV4 ranking problem.

Optional Hill saturation for an Hb carrier (disabled in template, Hb_capacity = 0).

Coupled simulation loop

1. Initialise C_tissue = 0; build lumen from cached hemodynamics
2. Rebuild lumen O_2 every N steps (or when band changes > tol)
3. Update C_tissue: explicit FD diffusion + M-M uptake + exchange
4. Clamp non-negative; check L_2 relative error vs previous frame
5. Stop on convergence, max time, or wall-clock cap

What the Solver Reports — and What It Doesn't

Reproducible outputs

Geometry: vessel mask, skeleton, radius map, distance-to-vessel CDFs

Hemodynamics: pressure field, speed map, reachability, transit-time proxy

Oxygen: tissue C (mM), tissue pO_2 (mmHg), lumen C, lumen pO_2

Scalars: min/median/mean tissue O_2, area fractions ≥ 15/20/30 mmHg

Files: *.npy arrays, summary CSV/JSON, full metrics log*

Parameter sweep (24 runs)

Parameter	Values	Role
P_wall (m/s)	1e-6, 3e-5, 1e-4	Wall exchange
Vmax (mM/s)	1e-5, 1e-4, 1e-3	Cell demand
Inlet pO_2 (mmHg)	80, 150	Perfusate
Thresholds (mmHg)	15, 20, 30	Op. cutoffs

Reported as area fractions, not viability.

All outputs are conditional on swept parameters. Scaffold permeability and seeded-cell metabolism are not measured here.

OXYSIM · PARAMETER SWEEP

Solver Stability Across the Parameter Sweep

12 runs · 3 swept parameters · zero non-physical flags

Chris Dargham

May 2026

What We Swept and Why

Three transport-level parameters

Parameter	Levels	Mechanism
P_wall (m/s)	1e−8, 1e−6	Vessel→tissue exchange
Vmax (mM/s)	1e−5, 1e−4, 1e−3	Tissue O_2 consumption
Inlet pO_2 (mmHg)	80, 150	Source oxygen tension

12 runs · 1 source image

Pre-flight sanity checks passed

0 runs with tissue pO_2 > inlet (concentration-order intact)

0 runs with tissue pO_2 > lumen max (no overshoot)

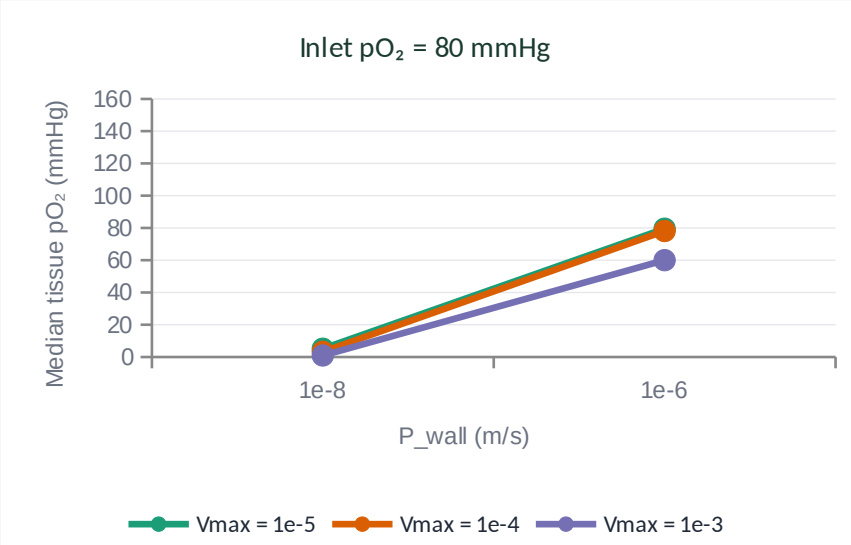
0 runs with negative concentrations (clamp working)

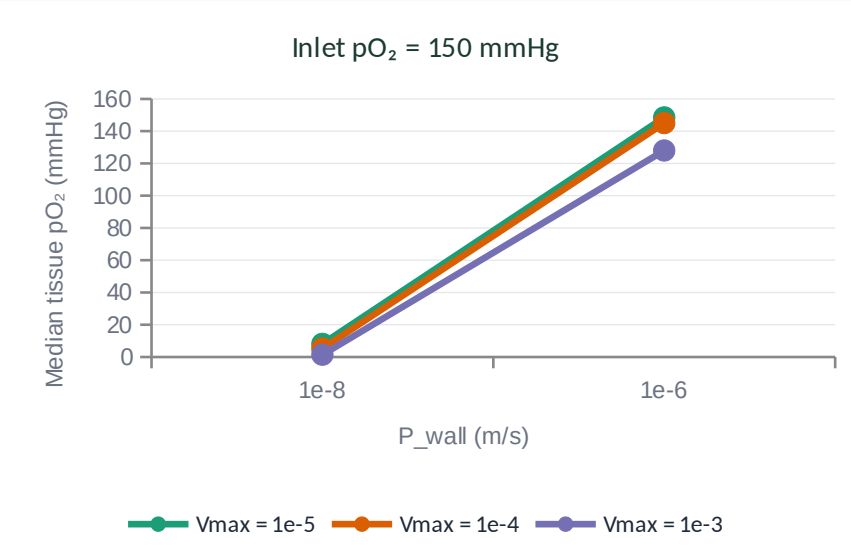
*Median tissue pO_2 ranged **0.91 → 149 mmHg** — wide, but bounded by inlet at every run.*

Sweep targets the three terms that govern tissue oxygenation in microvascular models (Secomb et al., 2004; Goldman & Popel, 2000).

MAIN RESULT

Wall Exchange Drives Tissue pO_2





Two-orders-of-magnitude jump in P_wall ⇒ ~20× rise in median tissue pO_2

Vmax shifts the curves only slightly; inlet pO_2 rescales them. P_wall is the dominant lever.

Threshold Response Is Binary

LOW EXCHANGE

P_wall = 1e−8 m/s

0%

of tissue ≥ 15 mmHg

median tissue pO_2 ≈ 4 mmHg

⇒

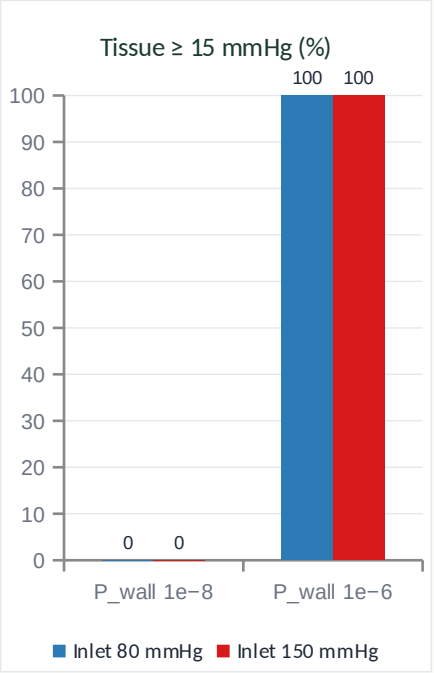
HIGH EXCHANGE

P_wall = 1e−6 m/s

100%

of tissue ≥ 15 mmHg

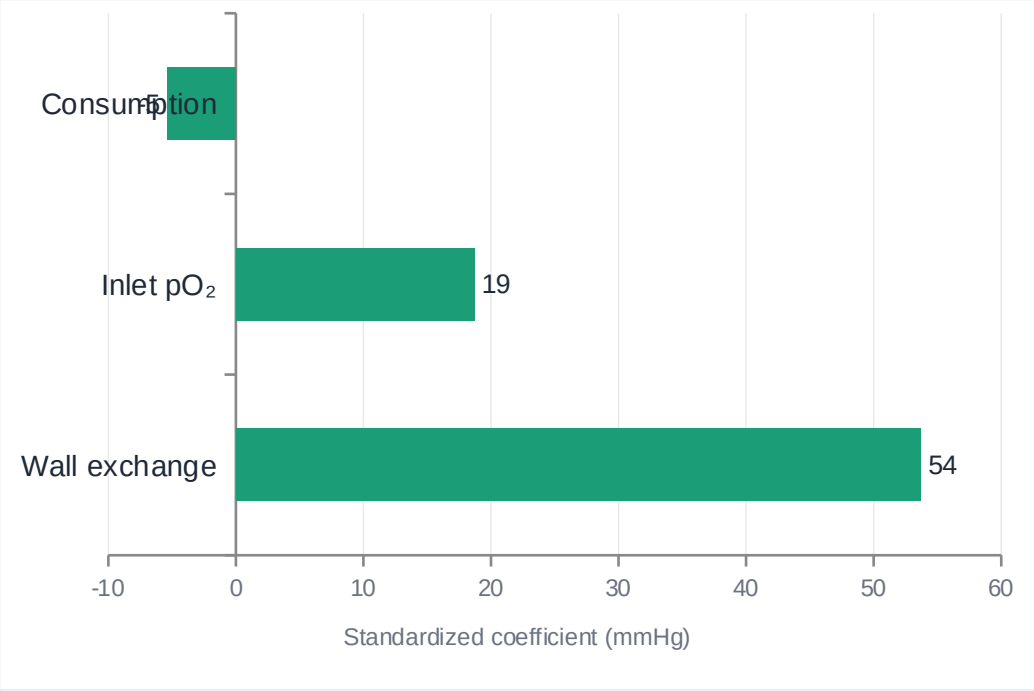
median tissue pO_2 ≈ 104 mmHg



Same step is observed for ≥ 20 and ≥ 30 mmHg thresholds — the model has a sharp exchange-controlled regime boundary.

STANDARDIZED COEFFICIENTS

A Linear Trend Model Confirms the Ranking



Model fit

$R^2 = 0.911$

adj $R^2 = 0.877$ · $n = 12$

Wall exchange $p < 10^{-4}$

Inlet pO_2 $p = 0.018$

Consumption $p = 0.42$ (n.s.)

Treated as a compact trend summary, not a predictive model.

CONCLUSION

The solver is stable and behaves as expected.

WHAT THIS SUPPORTS

- Numerical stability under controlled parameter variation
- Monotonic, physically interpretable response in P_wall, Vmax, inlet pO_2
- Distribution-wide shift (p10, median, p90 all move together)
- No tissue-over-inlet, tissue-over-lumen, or negative concentrations

WHAT THIS DOES NOT SUPPORT

- A measured value for plant-scaffold P_wall
- Cell-viability or tissue-survival predictions
- Generalization across scaffold geometries (n = 1 image)
- Any biological claim — outputs remain parameter-conditional

Next step: experimental calibration of the vessel-to-tissue exchange coefficient on a real decellularized scaffold.

Thank you.

CONCLUSION

A rigorous negative result, paired with a documented alternative.

Pipeline

Reproducible LeafVeinCNN → NetFlowV2 conversion of leaf venation into solver-ready hemodynamic graphs.

Failure analysis

Documented why GreensV4 is outside its reliable domain for reticulated plant networks — memory + ordering, both diagnosed.

OxySim prototype

Topology-compatible oxygen-screening solver with explicit assumptions, swept parameters, and stability audit.

Experimental calibration of plant-wall permeability and seeded-cell metabolism remains the next step before any biological viability claim.

Thank you.